Weekly Self-Study Tasks to Complete

Orchestration with Amazon EKS for Leaders

This document covers all of Week 8 of Phase 3 in the 4-hour Leaders model: 2 hours of instructor-led session, 1 hour of self-study video, and 1 hour of hands-on lab. The Core Resources table below fits inside the 1-hour self-study media budget. The Practice Activities section points to the Evergreen orchestration lab, where you will complete and validate Kubernetes manifests. Additional Content at the end is optional and sits outside the 4-hour budget.

Core Resources (~60 min)

Watch the specific resources called out below — not an entire course. Read the lab manifests alongside the videos so the concepts map to real YAML.

Topic	Material Description	Est. Time	Link / Reference
Kubernetes Big Picture	Kubernetes: Your First Project (Carlos Nunez, LinkedIn Learning, released Jul 28, 2025; 3h 48m course). Watch "Kubernetes at a Glance" (~25 min) — what Kubernetes is, the desired-state model, and the core objects (Pods, Deployments, Services).	~25 min	linkedin.com/learning — Kubernetes: Your First Project
Core Concepts	Kubernetes "Concepts" — Pods, Deployments, and Services (official docs). Read and skim how the desired-state model works.	~20 min	kubernetes.io/docs — Concepts
EKS	Amazon EKS "What is Amazon EKS?" (official docs) — read how EKS provides a managed Kubernetes control plane on AWS.	~15 min	docs.aws.amazon.com — EKS

Practice Activities (1 hour)

Complete this week's hands-on lab: "Evergreen Insurance Quote API — Orchestrating with Kubernetes on Amazon EKS." You will complete a Deployment and a Service, validate them, and (optionally) watch self-healing and scaling. The whole exercise takes about one hour and works without a live cluster.
In the lab you will:

- Open the Labfiles/evergreen-quote-eks folder and review the desired-state / self-healing model.
- Complete the Deployment: set replicas to 3, the container image, and the container port.
- Complete the Service: set the type to LoadBalancer, the label selector, and the target port.
- Validate the manifests with kubectl apply --dry-run=client -f k8s/.
- If a cluster is available, apply them, delete a pod, and watch Kubernetes self-heal.
- Scale the Deployment up and discuss replicas, rolling updates, and EKS vs portable Kubernetes.

The lab instructions (with "Why this matters" callouts throughout) are in the lab environment. Budget one hour and work at your own pace.

Additional Content (Optional)

The following resources are optional and recommended for students who want to go deeper beyond the core curriculum. They are outside the 4-hour weekly budget. This closes Phase 3, so use the extra material to connect orchestration back to the whole Evergreen journey.

Topic	Material Description	Link / Reference
Kubernetes	The Kubernetes documentation — the authoritative guide to Pods, Deployments, Services, scaling, and rollouts (Reference).	kubernetes.io/docs

Topic	Material Description	Link / Reference
Interactive	Kubernetes “Tutorials” with an in-browser playground — practice kubectl with a free cluster (Interactive).	kubernetes.io/docs/tutorials
Production	EKS Best Practices Guides — AWS guidance on security, reliability, and cost for production clusters (Reference).	aws.github.io/aws-eks-best-practices